



COURSE SLIDES

AWS Certified Machine Learning Engineer Associate Training

TGS-2024049340 | Hands-On Labs | Tertiary Infotech Academy Pte Ltd

Version 1

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About Tertiary Infotech Academy

WSQ course provider delivering ICT and business technology training.

UEN: 201200696W

Training format: instructor-led, hands-on, assessment-ready.

About the Trainer



Your Trainer

General Trainer template -
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS

PhD - specialises in AI, automation, software engineering and cloud learning.

DELIVERS

WSQ courses on cloud fundamentals, AI, automation and app development.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Digital Attendance

 Scan SSG digital attendance at required checkpoints.

 Minimum 75% attendance is required.

 Use the same registered learner identity throughout.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively - no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

5

Be punctual; return from breaks on time.

6

75% attendance is required.

Course Objectives

Data

Labs 1-2

Secure, ingest, transform and
validate ML data

Model

Labs 3-5

Select, train, tune, evaluate
and register

Deploy

Labs 6-7

Inference infrastructure and
MLOps automation

Operate

Lab 8

Monitoring, security, cost and
exam readiness

Learning Resources

 AWS Management Console: <https://console.aws.amazon.com>

 AWS documentation: <https://docs.aws.amazon.com>

 Official AWS Certified Machine Learning Engineer - Associate exam guide.

Assessment Flow Reminder



The TRAQOM QR code and LMS submission are both handled through the LMS/TMS portal.

Take the Online AWS MLA-C01 Practice Exams

- Reinforce AWS MLA-C01 domains with realistic, exam-style questions before assessment day.
- Use practice sets to test data preparation, model development, deployment, MLOps and monitoring.
- Review mode helps you check the correct answer and revisit the concept after each question.
- Complete the practice questions from Google Classroom, then bring difficult items to review.

Access and practise at:

[Google Classroom > Practice Exams](#)

The screenshot displays the 'AWS Machine Learning Engineer - Associate Practice Exams' page. At the top, there is a navigation bar with 'Tertiary Exams', 'Practice Exam', and 'About Us' links, along with a 'Get started' button. The main heading is 'AWS Machine Learning Engineer - Associate Practice Exams'. Below this, there are three interactive cards: 'MODE' with a 'Practice' button, 'FOCUS' with 'AWS MLA-C01' selected, and 'ACCESS' with a 'Class link' button. To the right, a 'PRACTICE DETAILS' box shows 'Exam code AWS MLA-C01', 'Use Review Domains 3', and a progress indicator.

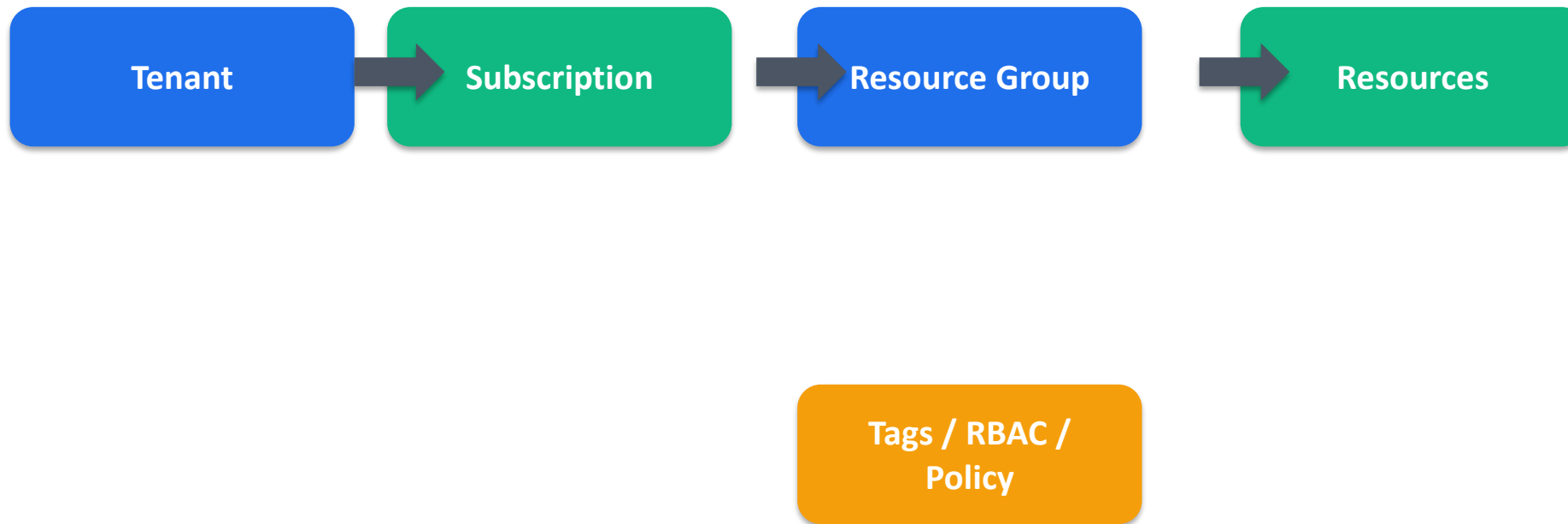
Two-Day Lesson Flow

 Day 1: Labs 1-4 - account safety, data preparation, modeling, training and tuning.

 Day 2: Labs 5-8 - evaluation, deployment, MLOps, monitoring, security and cost.

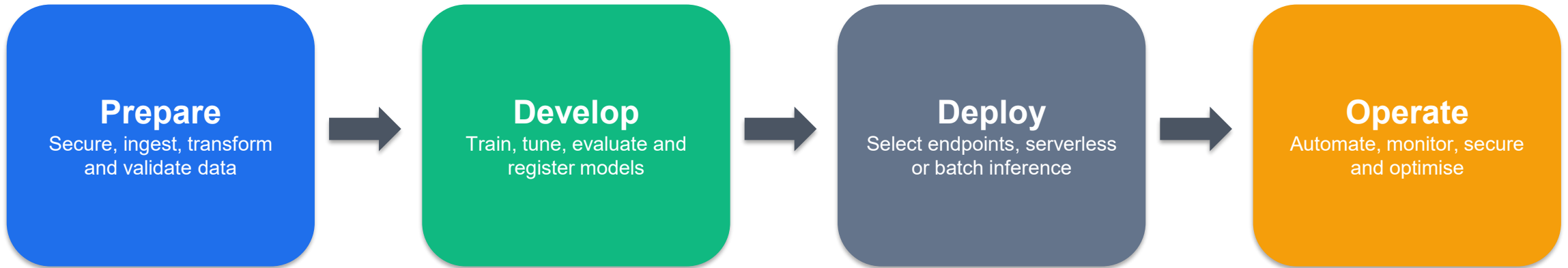
 Complete each validation and cleanup step before moving to the next lab.

AWS Resource Model



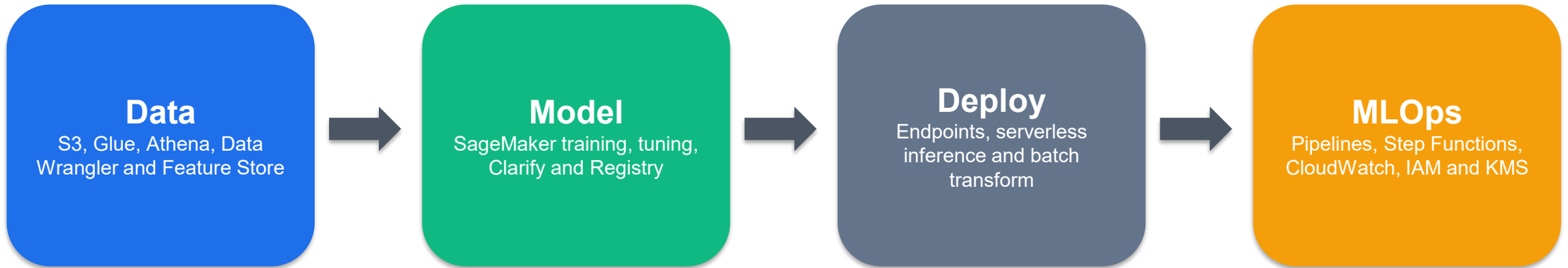
The labs repeatedly use this hierarchy: sign in to the tenant, work inside a subscription, place resources in tgs-aws-mla-labs, then govern them with tags, RBAC, locks, policy, cost controls and monitoring.

AWS ML Engineering Lab Flow



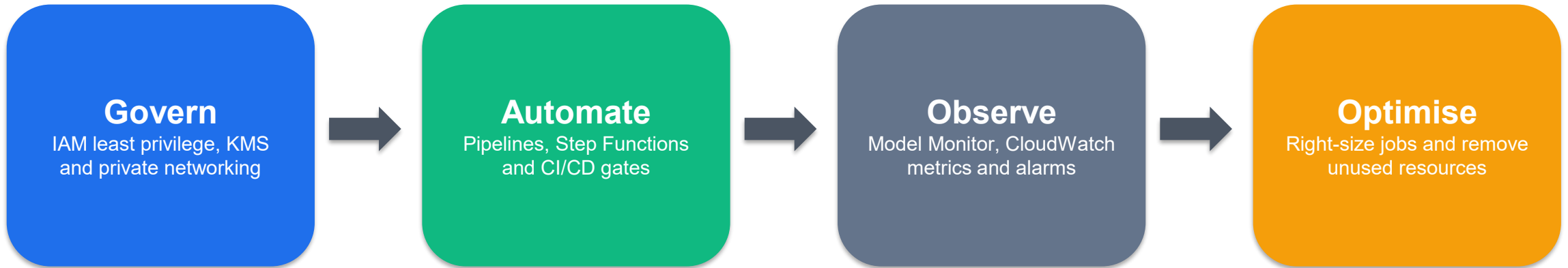
The slide deck, Learner Guide and Lesson Plan follow this same lab sequence.

AWS Services Practised



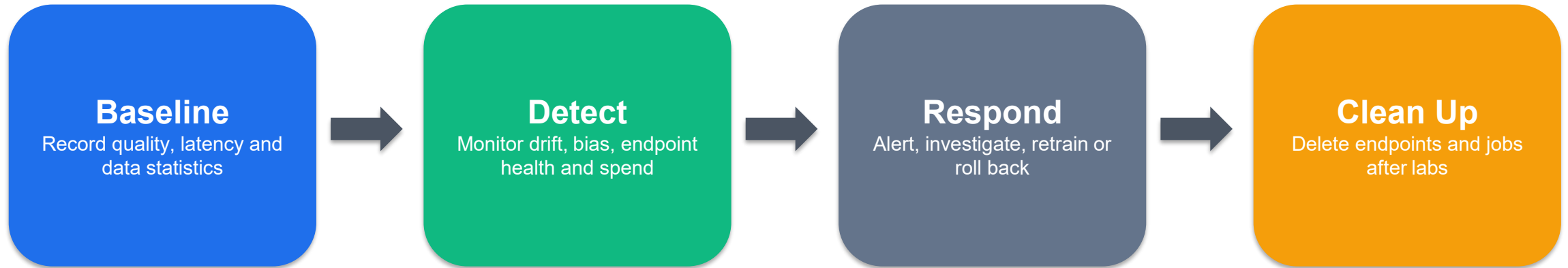
The service groups map directly to the eight labs and the MLA-C01 task domains.

Management and Governance Loop



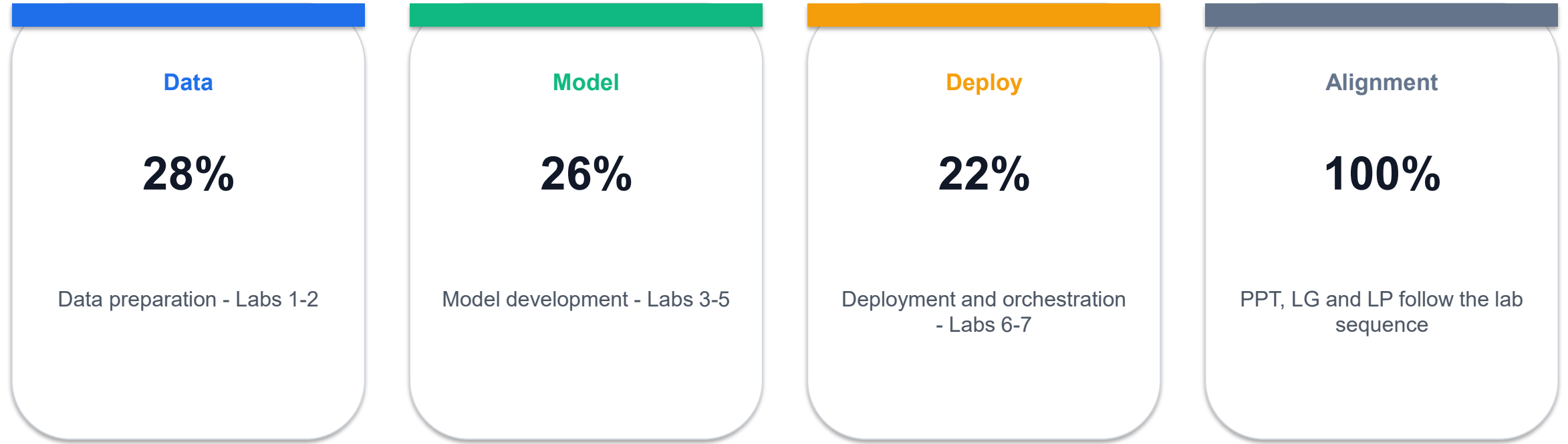
The Day 2 labs turn resources into governed cloud assets through identity, access, policy and monitoring.

Cost and Monitoring Lifecycle



Cost estimation, budgets, Advisor, Service Health, Monitor and cleanup are handled as one operating loop.

AWS MLA-C01 Domain Map

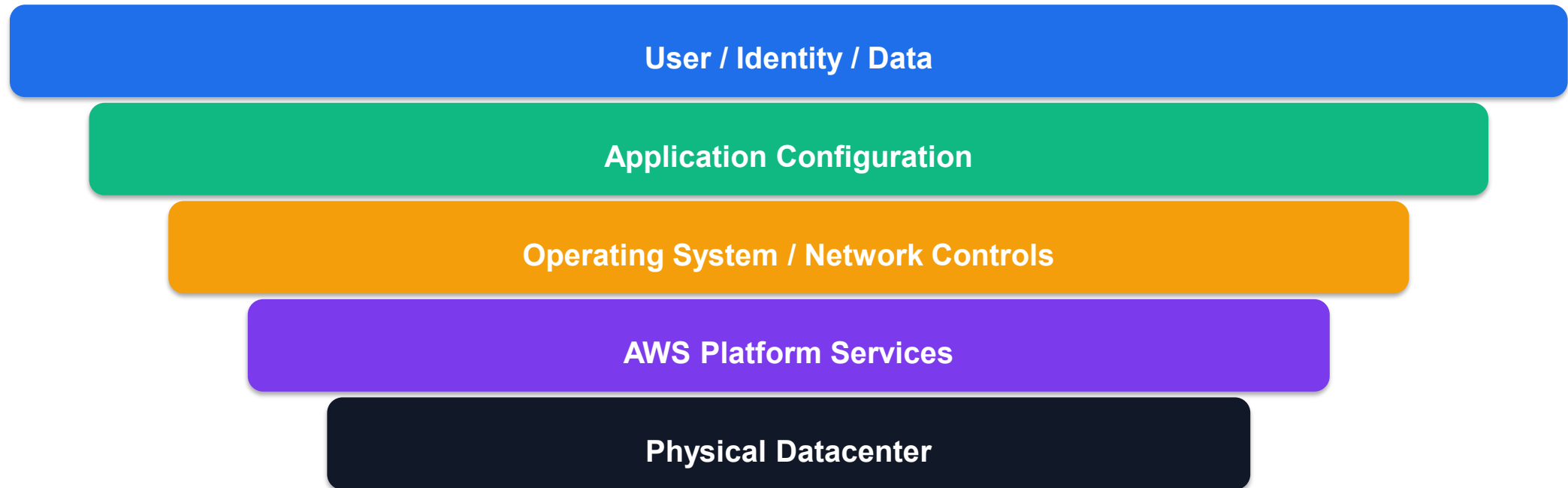


AWS MLA-C01 Domain Weight Matrix

Domain	Exam Weight	Labs	Practice Priority
Cloud Concepts	25-30%	1-3	Moderate
Architecture & Services	35-40%	4-11	High
Management & Governance	30-35%	12-16	High

This matrix mirrors the CySA+ risk-matrix layout, adapted to AWS MLA-C01 domain coverage and lab priorities.

AWS Shared Responsibility Layers



Learners configure the top layers in the labs; Microsoft operates the lower platform and physical layers.

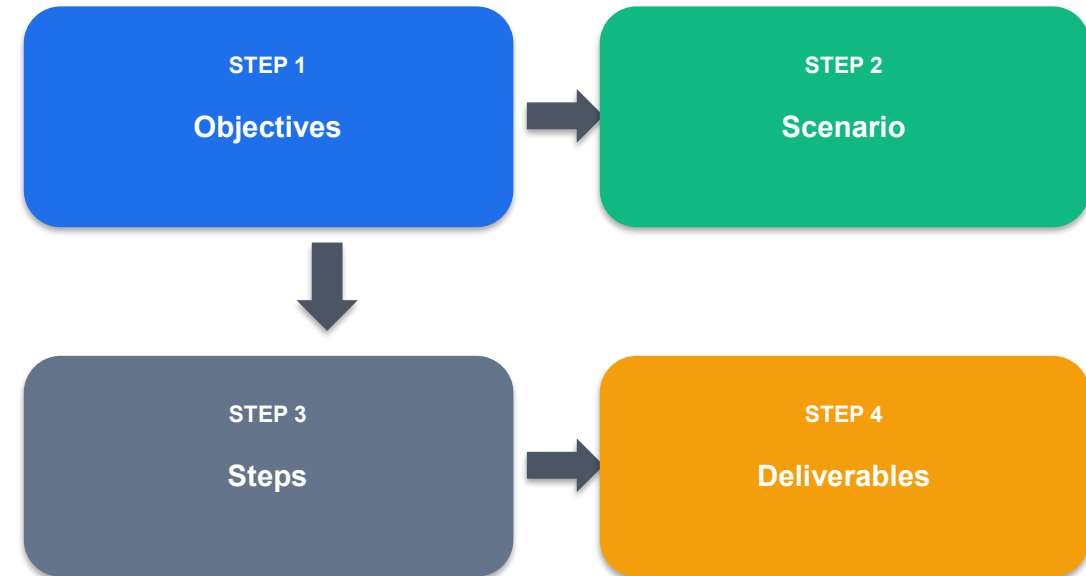
Lab 1 - Account Safety, IAM, S3 ML Data Lake, and Data Formats

Domain 1 - Data Preparation for Machine Learning

Day 1

labs/lab-01-account-
iam-s3-ml-data-
lake.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 1 - Key Steps



Objectives



Scenario



Steps



Deliverables



Checkpoint

Lab 1 - Verify and Clean Up

Verification

1. Complete the verification or What you learned section in the lab guide.
2. Run the cleanup section exactly as written.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Retain shared resources only when the next lab needs them.
3. Confirm no paid resource is left running.
4. Lab 8 performs the final course cleanup.

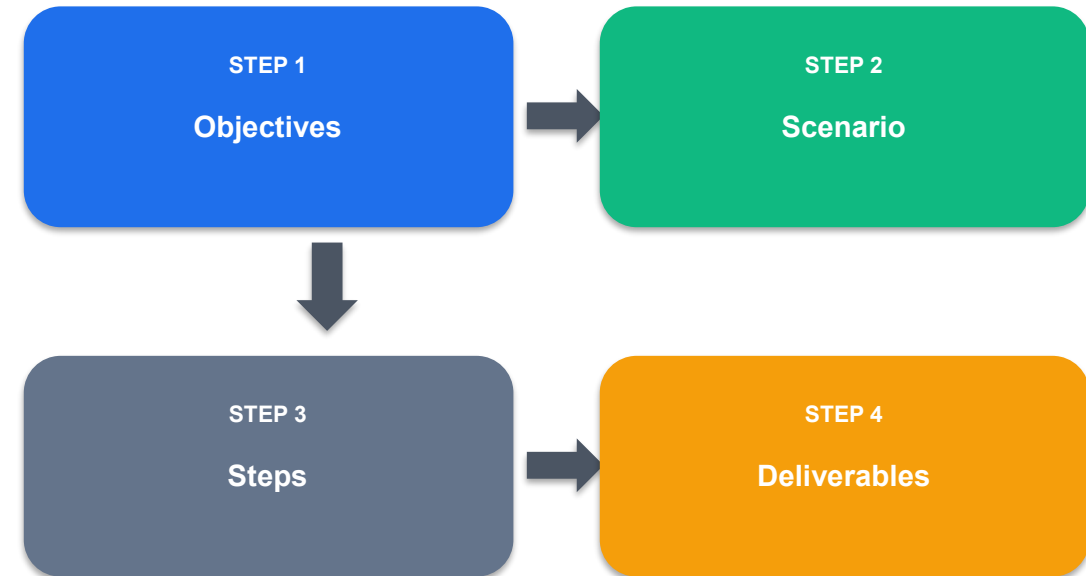
Lab 2 - Data Ingestion, Transformation, Feature Engineering, and Data Quality

Domain 1 - Data Preparation for Machine Learning

Day 1

labs/lab-02-data-
ingestion-feature-
engineering-
quality.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 2 - Key Steps

 Objectives

 Steps

 Checkpoint

 Scenario

 Deliverables

Lab 2 - Verify and Clean Up

Verification

1. Complete the verification or What you learned section in the lab guide.
2. Run the cleanup section exactly as written.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Retain shared resources only when the next lab needs them.
3. Confirm no paid resource is left running.
4. Lab 8 performs the final course cleanup.

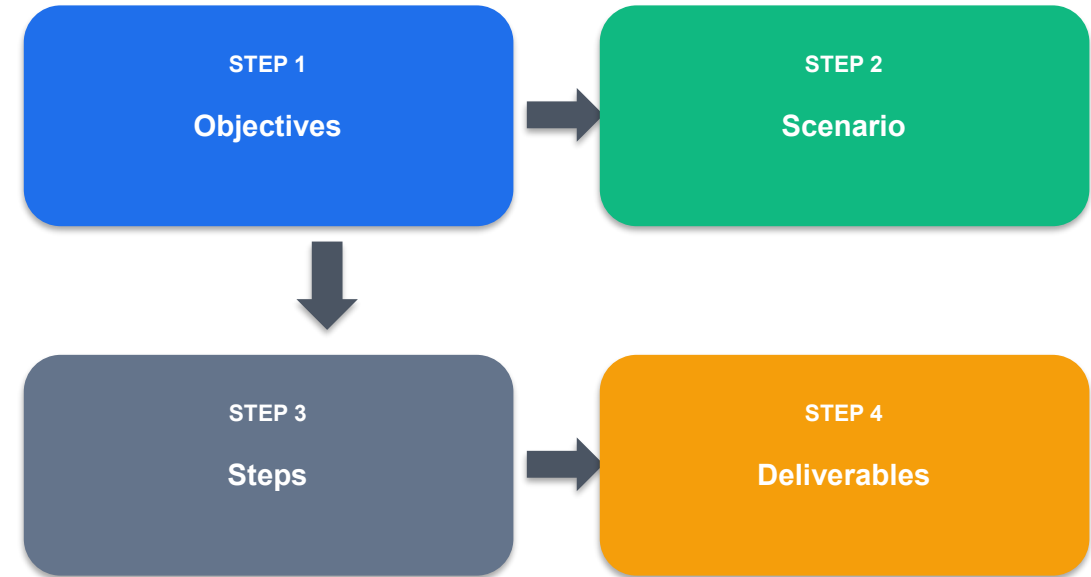
Lab 3 - Modeling Approach, SageMaker Algorithms, JumpStart, and AI Services

Domain 2 - ML Model Development

Day 1

labs/lab-03-
modeling-approach-
sagemaker-ai-
services.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 3 - Key Steps



Objectives



Scenario



Steps



Deliverables



Checkpoint

Lab 3 - Verify and Clean Up

Verification

1. Complete the verification or What you learned section in the lab guide.
2. Run the cleanup section exactly as written.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Retain shared resources only when the next lab needs them.
3. Confirm no paid resource is left running.
4. Lab 8 performs the final course cleanup.

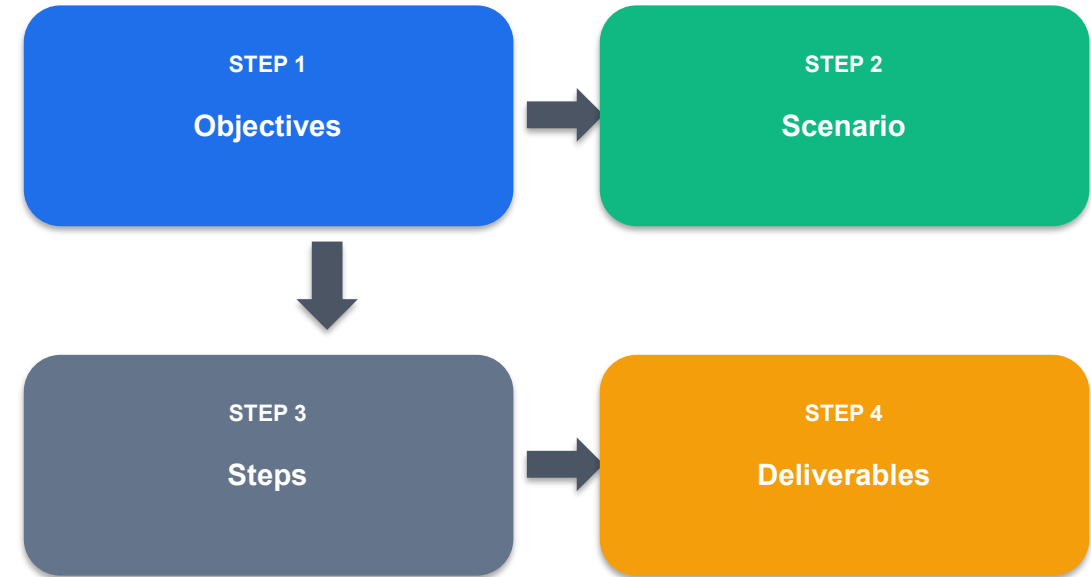
Lab 4 - Training, Hyperparameter Tuning, and Model Registry

Domain 2 - ML Model Development

Day 1

labs/lab-04-training-tuning-model-registry.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 4 - Key Steps

 Objectives

 Steps

 Checkpoint

 Scenario

 Deliverables

Lab 4 - Verify and Clean Up

Verification

1. Complete the verification or What you learned section in the lab guide.
2. Run the cleanup section exactly as written.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Retain shared resources only when the next lab needs them.
3. Confirm no paid resource is left running.
4. Lab 8 performs the final course cleanup.

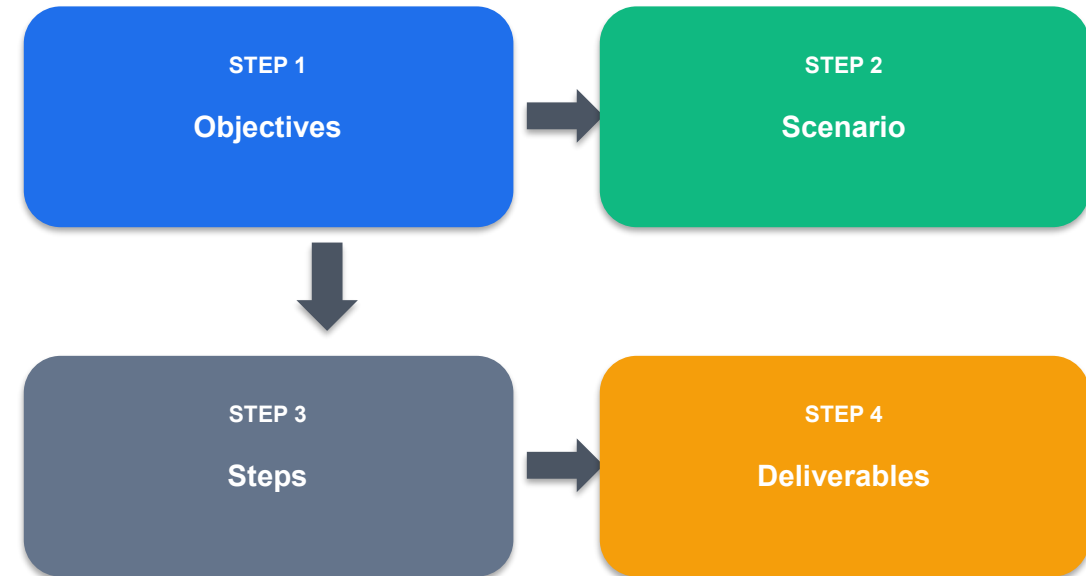
Lab 5 - Model Evaluation, Performance Metrics, and Bias Analysis

Domain 2 - ML Model Development

Day 2

labs/lab-05-model-
evaluation-
performance-
bias.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 5 - Key Steps



Objectives



Scenario



Steps



Deliverables



Checkpoint

Lab 5 - Verify and Clean Up

Verification

1. Complete the verification or What you learned section in the lab guide.
2. Run the cleanup section exactly as written.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Retain shared resources only when the next lab needs them.
3. Confirm no paid resource is left running.
4. Lab 8 performs the final course cleanup.

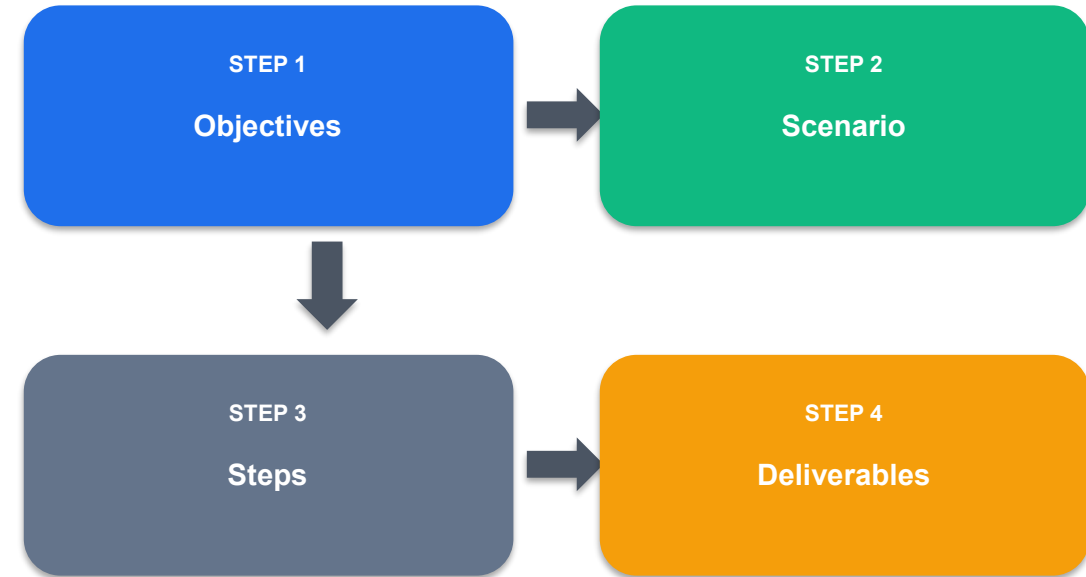
Lab 6 - Deployment Infrastructure, Endpoints, Batch Transform, and Auto Scaling

Domain 3 - Deployment and Orchestration of ML Workflows

Day 2

labs/lab-06-
deployment-
endpoints-inference-
infrastructure.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 6 - Key Steps

 Objectives

 Steps

 Checkpoint

 Scenario

 Deliverables

Lab 6 - Verify and Clean Up

Verification

1. Complete the verification or What you learned section in the lab guide.
2. Run the cleanup section exactly as written.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Retain shared resources only when the next lab needs them.
3. Confirm no paid resource is left running.
4. Lab 8 performs the final course cleanup.

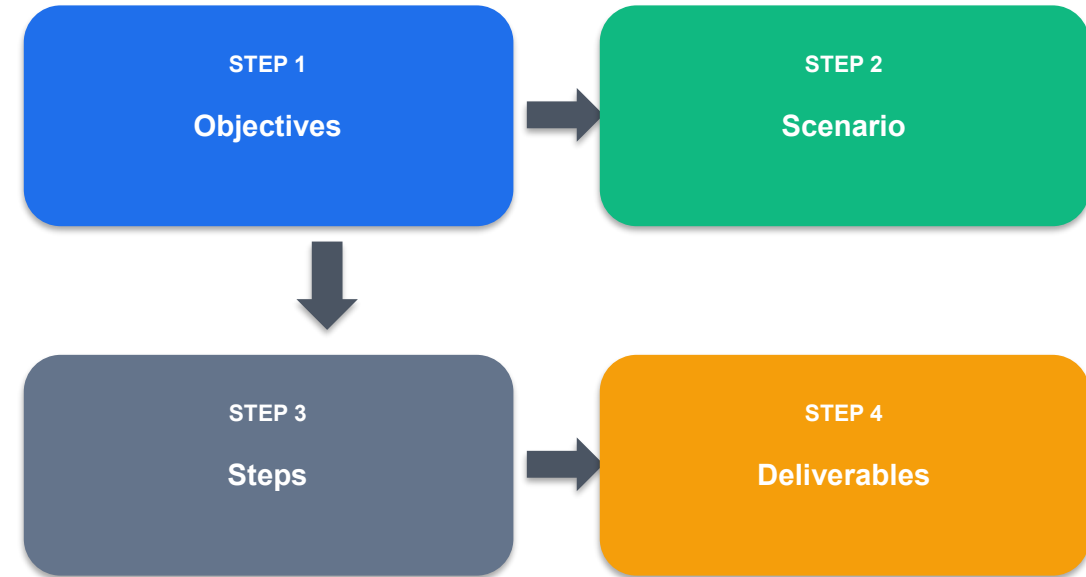
Lab 7 - MLOps Pipelines, CI/CD, and Workflow Orchestration

Domain 3 - Deployment and Orchestration of ML Workflows

Day 2

labs/lab-07-mlops-
pipelines-cicd-
orchestration.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 7 - Key Steps



Objectives



Scenario



Steps



Deliverables



Checkpoint

Lab 7 - Verify and Clean Up

Verification

1. Complete the verification or What you learned section in the lab guide.
2. Run the cleanup section exactly as written.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Retain shared resources only when the next lab needs them.
3. Confirm no paid resource is left running.
4. Lab 8 performs the final course cleanup.

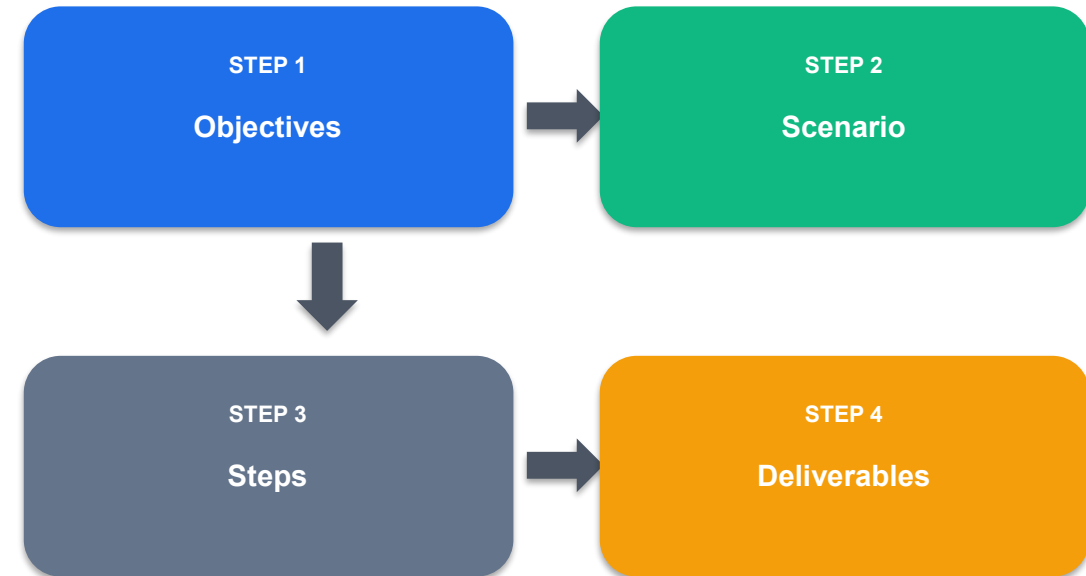
Lab 8 - Monitoring, Security, Cost Optimization, and Exam Review

Domain 3 - Deployment and Orchestration of ML Workflows

Day 2

labs/lab-08-
monitoring-security-
cost-exam-
review.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 8 - Key Steps



Objectives



Steps



Checkpoint



Scenario



Deliverables

Lab 8 - Verify and Clean Up

Verification

1. Complete the verification or What you learned section in the lab guide.
2. Run the cleanup section exactly as written.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Retain shared resources only when the next lab needs them.
3. Confirm no paid resource is left running.
4. Lab 8 performs the final course cleanup.

Course Wrap-Up

-  You prepared data and developed, evaluated, deployed and monitored ML workloads on AWS.
-  You practised MLOps, security and cost controls mapped to the MLA-C01 exam domains.
-  Complete the Lab 8 cleanup checklist to avoid ongoing AWS charges.